2. ELE 105.3 Basic Electrical Engineering (3-1-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

1. To analyze electric circuits (A.C. & D. C).
2. To work on electrical instrumentation projects.
3. To operate, distinguish and use electrical devices and machines.

Chapter	Content	Hrs.
1	**Introduction**	2
	Role of electricity in modern society, Energy sources and production, generation, transmission and distribution of electrical energy, consumption of electricity	
2	**DC Circuit Analysis**	15
	Circuits concepts (lumped and distributed parameters), linear and nonlinear parameter, passive and active circuits, Circuit elements (Resistance, capacitance and inductance), their properties and characteriscts in a geometrical and hardware aspects, color coding, Series of parallel compilation of resistances, Equivalent resistance and its calculation, star-delta transformation, concept of power, energy and its calculations, short and open circuit, ideal and non-ideal sources, source conversion, voltage divider and current divider formula, Kirchhoff's current and voltage laws, nodal method and mesh method of network analysis (without dependent source), network theorem (i.e Superposition, Thevonin's, Norton's), maximum power transfer.	
3	**Single Phase AC Circuits Analysis**	10
	Generation of EMF by electromagnetic induction, Generation of alternating voltage, sinusoidal functions-terminology (phase, phase angle, amplitude, frequency, peak to peak value), average values and RMS or effective value of any types of alternating voltage or current waveform, phase algebra, power triangle, impedance triangle, steady state response of circuits (RL, RC,RLC series and parallel) and concept about admittance, impedance reactance and its triangle), instantaneous power, average real power, reactive power, power factor and significance of power factor, resonance in series and parallel RLC circuit, bandwidth, effect of Q factor in resource.	
4	**Poly-phase AC Circuit Analysis**	6
	Concept of a balanced three phase supply, generation and differences between single phase over three phase system, star and delta connected supply and load circuits. Line and phase voltage\current relations, power measurement, concept of three phase power and its measurement by single and two wattmeter method	

5 Electric Machines **12**

Review of magnetic circuits

Transformers: Principle of operations, features, equivalent circuits, efficiency & regulation, open circuit & short circuit tests

DC motors: Performance & operation, basic characteristics of motors & generators, speed control & selection of motors

AC machines: Induction motors (working principles, construction features and uses), Synchronous motors (working principles, construction and uses)

Textbook

1. Boylested, Albert , "Introduction of Electric circuit" Prentice Hall of India Private Limited, New Delhi
2. Tiwari, S.N, "A first course of electrical engineering" att. Wheeler & Co.Ltd Allabhad.

References:

1) Thereja B. L & Thereja A. K " A text book of Electrical Technology, S Chand Publication.
2) Jain& Jain" ABC of Electrical Engineering""

Laboratory Work:

1. To measure current, voltage and power across the passive components.
2. To verify Kirchhoff's Current Law (KCL) & Kirchhoff's Voltage Law (KVL)
3. To verify Thevenin's Theorem.
4. To verify maximum power transfer theorem.
5. To verify superposition theorem.
6. To measure three phase power by using two wattmeter
7. To determine efficiency and voltage regulation of a single-phase transformer by direct loading.
8. To study open circuits & short circuits tests on a single phase transformer
9. To study the speed control of dc shunt motor by.
   i. Varying the field current with armature voltage held constant field control.
   ii. Varying the armature voltage with field current held constant armature control.
10. To study open circuits and load test on a dc shunt generator (separately excited)
    i. To determine magnetization characteristics
    ii. To determine V-I characteristics of a dc shunt generator

3. CHM 103.4 Chemistry (4-1-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

1. Analyze chemical behavior of materials
2. Analyze the water quality.
3. Analyze environmental aspects of various elements and compounds.

Chapter	Content	Hrs.
1.	**Ionic Equilibria and Electro Chemistry** Introduction and types of buffer solutions, mechanism, Henderson- Hassel Balch equation, electro chemical cells, Galvanic cell, cell notation, cell reaction, cell potential, single electrode potential , standard electrode potential, electro chemical series & its applications, Nernst equation, corrosion, its type, mechanism and control.	**10**
2	**General Inorganic Chemistry** Ionization energy, electro negativity, electron affinity, characteristic properties of S and P block elements, introduction of Transition metals, characteristic properties of transition metals (electronic configuration, atomic radii, variable oxidation states, complex formation, color and magnetic properties.	**10**
3	**General Organic Chemistry** Reaction intermediates: carbocations, carbanions and carbon free radicals, stereoisomerism Organic reaction mechanism: Nucleophilic substitution reactions (SN1&SN2), Electrophilic aromatic substitution, Elimination (E1 & E2), Electrophilic and free radical addition reaction	**10**
4	**Polymer Chemistry** Polymer and polymerization, basic concepts , types of polymerization (addition and condensation), thermoplastics and thermosetting plastics, preparation, properties and uses of : polyethylene, PVC, Teflon, Bakelite's, Nylon, polyester, polyurethelene and silicon, rubber, processing of natural rubber and vulcanization	**10**
5	**Analytical Chemistry** Introduction and application of following analytical techniques: fractional distillation, chromatography (paper, thin layer) NRM, Mass spectroscopy	**6**
6	**Industrial Chemistry** Introduction of paints, chemistry of paints, lubricants and its classification, cement, chemistry of cement, manufacture & its setting mechanism, Explosives: TNT, TNG	**4**
7	**Environmental Chemistry** Water pollution- causes of water pollution, acid rain, alkalinity COD, DO, Hardness,(effects to human	**10**

health), control

Air pollution: causes, global warming and climate change ozone layer depletion and CFC, control measures

Soil pollution: causes, effects and its control measures

Laboratory works
Objectives

- Use techniques apparatus and instructions properly
- Interpret, evaluate and report upon observations and experimental results
- Design/plan on investigation, select techniques, apparatus and materials
- Evaluate methods and suggest possible improvements

Laboratory works

1. Determine of total alkalinity of given water sample
2. Determination of hardness of water sample by complexometric titration
3. Determination of free chorine in the given water sample
4. Preparation of buffer and determination of pH of the solution
5. Estimation of DO in the given water sample
6. Estuation of COD in the given water sample
7. To separate the ink mixture by paper chromatography or TLC (Demo)
8. To purify a sample of mixture of crude alcohol and petroleum by fractional distillation (Demo)
9. To estimate carbon monoxide gas in the car exhaust (Demo)

Text books:

1. Physical Chemistry, B.S. Bahl and G.D. Tuli
2. Advanced inorganic Chemistry, J. D. Lee
3. Advance Organic Chemistry, Morisson and Boyd 6th edition
4. Engineering Chemistry (with experiments), Sunita Rattan , 4th edition Publisher of Engineering and Computer books

Reference Books:

1. Satya Prakash, Tuli, Basu, Madan: Advanced Inorganic Chemistry, S. Chanda & Company Ltd, New Delhi
2. Polymer Science, V.R. Gowariker, N. V. Vishwanathan
3. Environmental Chemistry, Anil Kumar Datta
4. Advanced Organic Chemistry, A. Bahl and B. S. Bahl
5. Text book of Chemistry P.N. Chaudhary and M.L. Bhusal
6. Lab manual of Engineering Chemistry by S.K. Bhasin and Sudha Rani

4. ENG 104.2 Communication Technique (2-2-1)

Evaluation:

	Theory	Practical	Total
Sessional	50	-	50
Final	50	-	50
Total	100	-	100

Course Objectives:
The main objectives of this course are:

4. To develop the ability to deliver technical knowledge orally in English.
5. To be able to comprehend and take notes after listening.
6. To fasten reading skills in technical and non-technical reading materials.
7. To develop summarizing skills in writings.
8. To write reports, letters, description on technical talks, seminar papers, memoranda, application

Chapters	Content	Hrs.
1	**Review of Written English**	2

- Identification of Sentence and clause
- Classification of sentence (simple, compound, complex)
- transformation of sentences

2 Oral Communication and Note Taking — 9

- Variety of English (BrE, AmE, formal, informal, polite, familiar, tentative)
- General rules of pronunciation (English Vowels and Consonants)
- General rules of stress and intonations
- Oral presentation/technical talk: Environmental pollution, construction, water resources, impact of satellite communication, urban development, impact of computer in modern society

3 Technical Writing Skills — 10

- Preparation of short memoranda (Importance, formats)
- Business letters (Importance-purposes)
- Preparation of job application and CV
- Description writing (Process, Mechanism, Place etc.)
- Calling meeting and writing minutes, notification, preparation of agenda

4 Reading Skills — 9

- Comprehension questions and exercises from:
- The use and the misuse of science, Road foundation, Beauty, Custom, The story of an hour (Kate Chopin), Knowledge and wisdom, Freedom, Letter from foreign grave (D. B Gurung), Natural Resources of Nepal: Forests & Water (Mani Bhadra Gautam)
- Note making and precise writing from any passage

Tutorial Works:

1. Some general rules of pronunciation..
2. To present a seminar paper/report/proposal.
3. To participate in a group discussion.
4. To conduct a meeting.
5. To prepare and practice to face an interview.

Textbook:

1. Andrea J. Rutherford. *Basic Communication Skills for Technology*. 2nd Edition. Addision Wesley. Pearson Education Asia (LPE) ISBN: 8178082810
2. Khanal Arjun, Communication Skills in English, Sukunda Pustak Bhawan, 2010

Reference Books:

1. Anne Eisenberg, *Effective Technical Communication*, Mc-Graw Hill 1982.
2. V.R. Narayanaswami, *Strengthen your writing*, Orient Longman, Madras.
3. Champa Tickoo & Jaya Sasikumar, *Writing with a Purpose*, Oxford University Press, Bombay.
4. A handbook of pronunciation of English words (with 90-minute audio cassettes) Communication Skills in English.
5. Chopin, Kate. "The Story of an Hour", *Creative Delights*
6. *Gautam Shreedhar, Creation & Criticism: A Miscellaneous Thought*
7. *Gautam Mani Bhadra, Essays, Stories, Passages, Paragraphs and Letter writing for the Young Learners, Nirantar Prakashan, Kathmandu, 2008*

5. MTH 111.3 Engineering Mathematics I (3-2-0)

Evaluation:

	Theory	Practical	Total
Sessional	50	-	50
Final	50	-	50
Total	100	-	100

Course Objectives:

After the completion of this course students will be able to apply the concept of calculus (Differential and integral), analytical geometry and vector in their professional courses.

Chapter	Content	Hrs.
1	**Limit, Continuity and Derivative:**	15

i. Limit, continuity and Derivative of a function with their properties
ii. Mean values Theorem with their application
iii. Higher order derivative
iv. Indeterminate forms
v. Asymptote
vi. Curvature
vii. Ideas of curve tracing
viii. Extreme values of functions of single variables

2	**Integration with its Application:**	17

i. Basic integration, standard integral, definite integral with their properties
ii. Fundamental theorem of integral calculus (without proof)
iii. Improper integral
iv. Reduction formulae and use of beta Gamma functions
v. Area bounded by curves
vi. Approximate area by Simpsons and Trapezoidal rule,
vii. Volume of solid revolution

3	**Two dimensional geometry:**	7

i. Review (circle, Translation and rotation of axes)
ii. Conic section(parabola, ellipse, hyperbola),
iii. Central conics (Introduction only).

4.	**Vector Algebra:**	6

i. Review of vector and scalar quantity
ii. Space coordinates
iii. Product of two or more vectors
iv. Reciprocal system of vectors and their properties
v. Equations of lines and planes by vector methods

Text Books:

1. Engineering Mathematics I: Prof. D.D Sharma (Regmi), Toya Narayan Paudel, Hari Prasad Adhikari, Sukunda Publication Bhotahity , Kathmandu
2. Calculus and analytical geometry: George B Thomas, Ross L. Finney

Reference Books:

1. Calculus with analytical geometry: E.W. Swokoswski.
2. Coordinate Geometry: Lalji Prasad.
3. Vector Analysis: M. B. Singh
4. Integral Calculus: G.D. Panta.

8. MEC 178.1 Mechanical Workshop (0-0-3)

Evaluation:

	Theory	Practical	Total
Sessional	-	100	100
Final	-	-	-
Total	-	100	100

Course Objectives:

To provide instructions and practical experience in basic mechanical workshop methods

Course Contents:

Chapters	Content	Hrs.
1	**Mechanical Workshop Materials** Introduction to mechanical workshop, Basics of steel and cutting materials, Common non-ferrous metals, Important mechanical properties.	4
2	**Measurement and Measuring Equipment**	1.5
3	**Bench Tools and Basic Hand Operations** Filing, Sawing, Sheet metal working, screw thread and screw thread cutting	1.5
4	**Joining Processes** Riveting, Soldering, Brazing, Welding	1.5
5	**Introduction to Machine Tools** Elements of machine tools, Cutting actions and tooling	1.5
6	**Familiarization with Basic Machine Tools** Lathe, Milling machine, Drill presses, Power saws, Shaping Machine and Grinding machines	5

Practical:

6. To convert a metallic job piece into a prescribed form using mechanical bench tool.
7. To turn a cylindrical job piece to prescribed dimension by using lathe machine.
8. To convert a metallic job piece to prescribed dimension by using milling machine.
9. To provide surface finish to a metallic piece by using the shaper machine.
10. To weld required metallic pieces together by using electric arc and gas welding, to given shape and size.
11. To make knot & bolt of given size and type
12. To make tray/dust bin/ pen holder or similar item with sheet metal.

Reference Books:

11. Anderson and E.E. Tatro, *Shop Theory*, McGraw-Hill 5th edition, 1942.
12. Lascoe, C.A. Nelson and H.W. Porter, *Machine Shop Operation and Setups*, American Technical Society, 1973.

13. *Machine Shop Practice – Volume II*, Industrial Press, New York, 1971.
14. Oswald, *Technology for Machine Tools*, McGraw-Hill Ryerson, 3rd edition.
15. Oberg, Jones and Gorton, *Machinery's Handbook*, 23rd edition, Industrial Press, New York

9. CMP 103.3 Programming in C (3-0-3)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

The object of this course is to acquaint the students with the basic principles of programming and development of software systems. It encompasses the use of programming systems to achieve specified goals, identification of useful programming abstractions or paradigms, the development of formal models of programs, the formalization of programming language semantics, the specification of program, the verification of programs, etc. the thrust is to identify and clarify concepts that apply in many programming contexts:

Chapter	Content	Hrs.
1	**Introduction** History of computing and computers, programming, block diagram of computer, generation of computer, types of computer, software, Programming Languages, Traditional and structured programming concept	3
2	**Programming logic** Problems solving(understanding of problems, feasibility and requirement analysis) Design (flow Chart & Algorithm), program coding (execution, translator), testing and debugging, Implementation, evaluation and Maintenance of programs, documentation	5
3	**Variables and data types** Constants and variables, Variable declaration, Variable Types, Simple input/output function, Operators	3
4	**Control Structures** Introduction, types of control statements- sequential, branching- if, else, else-if and switch statements, case, break and continue statements; looping- for loop, while loop, do—while loop, nested loop, goto statement	6
5	**Arrays and Strings** Introduction to arrays, initialization of arrays, multidimensional arrays, String, function related to the strings	6
6	**Functions** Introduction, returning a value from a function, sending a value to a function, Arguments, parsing arrays and structure, External variables, storage classes, pre-processor directives, C libraries, macros, header files and prototyping	6

7 Pointers **7**

Definition pointers for arrays, returning multiple values form functions using pointers. Pointer arithmetic, pointer for strings, double indirection, pointer to arrays, Memory allocation-malloc and calloc

8 Structure and Unions **5**

Definition of Structure, Nested type Structure, Arrays of Structure, Structure and Pointers, Unions, self-referential structure

9 Files and File Handling **4**

Operating a file in different modes (Real, Write, Append), Creating a file in different modes (Read, Write, Append)

Laboratory:

Laboratory work at an initial stage will emphasize on the verification of programming concepts learned in class and use of loops, functions, pointers, structures and unions. Final project of 10 hours will be assigned to the students which will help students to put together most of the programming concepts developed in earlier exercises.

Textbooks:

1. Programming with C, Byran Gottfried
2. C Programming, Balagurusami

References

1. A book on C by A L Kely and Ira Pohl
2. The C Programming Language by Kerighan, Brain and Dennis Ritchie
3. Depth in C, Shreevastav